

Alvaro Perdomo Gutierrez

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with **12+** years across transportation, healthcare, and e-commerce, building reliable analytics platforms and ETL pipelines. Experienced in data warehousing, transformation logic, and pipeline orchestration using SQL, Airflow, and Databricks. Skilled in data quality, testing, and documentation practices to deliver trusted data products for analytics and business intelligence.

SKILLS

Languages: SQL, Python, JavaScript, TypeScript, HTML, CSS

Data & ML: Airflow, Databricks, dbt, Spark, Pandas, NumPy, TensorFlow, MLflow

Backend: Node.js, FastAPI, Express, NestJS, Spring Boot

Database: PostgreSQL, Snowflake, BigQuery, TcaplusDB, MongoDB, Redis, Elasticsearch

Testing: Pytest, Jest, Playwright, Cypress, k6

Cloud & DevOps: AWS, GCP, Azure, Kubernetes, Docker, Terraform, GitHub Actions, Jenkins, EC2, S3, CloudFront, RDS, Lambda

Others: RESTful API, GraphQL, gRPC, Kafka, RabbitMQ, Git, GitHub, Jira, Confluence, Grafana, Prometheus, Datadog, Git, Agile, Scrum

EXPERIENCE

Leanware

May 2024–Present

Senior Software Engineer

- Spearheaded the design and implementation of an analytics dashboard for a transportation management platform, building data ingestion pipelines using **Python**, **SQL**, and **Airflow** to aggregate real-time traffic and routing metrics from multiple source systems into a unified data warehouse.
- Engineered analytics-ready datasets and transformation logic in **Databricks** using **SQL** and **dbt**, enabling stakeholders to query 18+ months of historical traffic patterns and vehicle performance data with sub-second query latency for executive reporting.
- Collaborated with analytics and operations teams to define data quality checks, reconciliation logic, and monitoring rules, implementing automated validation tests that caught 94% of data anomalies before they reached downstream dashboards and prevented two critical reporting delays.
- Optimized a slow-running ETL pipeline by refactoring transformation queries and implementing incremental load strategies with **Airflow**, reducing daily processing time from 2.5 hours to 34 minutes and enabling same-day data availability for demand-forecasting analytics.

- Documented data lineage, transformation assumptions, and business logic for the transportation analytics platform in **Confluence**, ensuring that new team members could understand and modify pipelines independently within their first two weeks.
- Participated in code reviews and peer feedback sessions, establishing naming conventions and SQL formatting standards that improved pipeline readability and reduced merge conflicts during team collaboration on shared **GitHub** repositories.
- Mentored a junior analyst on dbt project structure and incremental loading patterns, helping them ship two self-service semantic layer tables that eliminated 12+ manual reporting queries per week.
- Supported the migration of legacy data processes from batch scripts to **Kafka** event-driven workflows, eliminating 8-hour processing windows and enabling near-real-time operational metrics for the logistics team.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Led the consolidation of disparate inventory data sources into a unified **Databricks** lakehouse using **Airflow**, **SQL**, and **dbt**, enabling the analytics team to build 40+ new KPI dashboards on top of a single source of truth for 172M+ apparel items and reducing data discrepancies by **89%**.
- Redesigned the garment measurement and resale-pricing ETL pipeline to process inventory changes in real-time using **Kafka** topic subscriptions, reducing pricing staleness from 6 hours to under 5 minutes and contributing to a **2.8%** lift in iOS conversion rates.
- Built comprehensive data quality and reconciliation logic in **Python** using **Pandas** and **NumPy**, executing daily validation checks across inventory, pricing, and image metadata feeds, which surfaced a critical supply-chain data gap affecting 8,000+ product listings.
- Partnered with the ML and analytics teams to document feature stores and training datasets, establishing metadata standards and data lineage tracking in **Confluence** that accelerated model deployment cycles from 2 months to 2 weeks.
- Optimized **Snowflake** query performance by analyzing table scan patterns and implementing dynamic clustering on high-cardinality columns, achieving a **31%** reduction in query execution time for the company's largest analytical queries and lowering cloud costs by **\$18K** monthly.
- Supported the testing and validation of a visual recognition ML model trained on 275K+ product images, designing data sampling strategies and performance benchmarking queries that confirmed model accuracy across apparel categories before production release.
- Mentored two junior engineers on SQL fundamentals and ETL design patterns, conducting weekly office hours and pair-programming sessions that prepared them to independently own new pipeline components within three months.
- Collaborated with BI teams to translate business requirements into semantic layer definitions, building reusable **dbt** models and dimension tables that reduced dashboard query complexity and improved analyst productivity by **40%**.

Spring Health

Software Engineer

Nov 2020–Mar 2022

- Developed core ETL pipelines for the Care Navigation platform using **SQL** and **Python**, ingesting member eligibility, benefits, and clinical data from 15+ enterprise customers while preserving data integrity and HIPAA compliance during a **6x** growth period.
- Built data validation and reconciliation logic in **Airflow** to match 2M+ member records across legacy and new systems, ensuring zero duplicate enrollments and accurate dependent coverage attribution during the global family launch in September 2021.
- Engineered a **Spark**-based batch job to process jurisdiction-aware matching rules and localization transformations for Fortune 500 onboarding, enabling the platform to scale from 4 US states to 12 countries while maintaining clinical validity of care recommendations.
- Designed and implemented operational data marts in **PostgreSQL** that powered dropout-risk prediction and demand-forecasting models, allowing the clinical team to proactively identify at-risk members and allocate resources more effectively across regions.
- Contributed to the migration and optimization of analytics queries from legacy systems to a cloud data warehouse, reducing report generation time from 45 minutes to under 8 minutes and enabling real-time dashboard refreshes for executive stakeholders.
- Participated in data quality audits and root-cause analysis sessions following production incidents, documenting findings and recommending improvements to monitoring and validation logic that prevented similar issues in downstream releases.
- Collaborated with product and analytics teams to understand requirements for new reporting capabilities, translating high-level requests into data transformations and curated tables that enabled 18 new health-plan analytics dashboards.

Tencent Games

Software Engineer

Nov 2014–Nov 2020

- Built telemetry and analytics pipelines for the Honor of Kings mobile game, ingesting event data from 100M+ daily active users using **Kafka**, processing logs in **Spark**, and storing aggregated metrics in **TcaplusDB** to power real-time dashboards and post-game analytics.
- Developed **SQL** transformation logic and aggregation queries to compute player behavior cohorts, retention metrics, and monetization KPIs, enabling the game analytics team to segment users and measure the impact of gameplay changes and seasonal content updates.
- Implemented data validation and quality monitoring for game telemetry pipelines, configuring alerting rules in **Prometheus** and building diagnostic dashboards in **Grafana** that detected data quality issues within minutes and reduced debugging time by **35%**.
- Contributed to the design and deployment of an anti-cheat infrastructure that aggregated behavioral signals, device metadata, and player reports into a **PostgreSQL** database, enabling the security team to flag and investigate suspicious accounts in near-real-time.
- Collaborated with product and game design teams to translate feature requirements into event instrumentation plans, documenting data schemas and transformation assumptions that ensured analytics reliability across 18-month seasonal content roadmaps.

- Executed backlog items to optimize game telemetry storage and reduce infrastructure costs, including partitioning strategies for time-series data and archival policies that lowered database storage costs by **22%** while maintaining query performance.
- Supported the onboarding of a new game title to the shared analytics platform by validating event schemas, testing data pipelines, and documenting run-books, enabling the new game team to launch analytics dashboards within two weeks of production release.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014